PH 113
Physics

Nuclear Shell Model

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Nuclear Shell Model

- **Definition**

In nuclear physics, the **nuclear shell model** is a theoretical model to describe the **atomic nucleus (in terms of energy levels).**

- The nuclear shell model was proposed by **Dmitry Ivanenko** in 1932 and further developed independently by several physicists such as Maria Goeppert-Mayer, Eugene Paul Wigner and J. Hans D. Jensen in 1949.
- It must be noted this model is based on the Pauli exclusion principle to describe the structure of the nucleus in terms of **energy levels**.

Features of Shell Model

The important features of nuclear shell model are:

- The Shell Model is partly analogous to atomic shell model which describes the arrangements of electrons in an atom.
- The nucleons move randomly in a nucleus and collide into each other frequently in liquid drop model. The **shell model** suggests that each nucleon in a nucleus moves in a well defined orbit and hardly makes any collision. This is why this model is also called as **independent model.**.

Features of Shell Model

- As Nuclear Shell Model is analogous to atomic shell model so filled shells results in greater stability
- The nucleons in a nucleus obey **Pauli exclusion principle(no two nucleons may occupy same state at the same time).** The neutrons and protons are treated separately when their states are considered . Each have its own array of available quantized states.
- In this model each nucleon is assume to exist in shell just like in atomic model.
- The nuclei shell are associated with certain **Magic Numbers.**

Features of Shell Model

❑ Magic Number

In nuclear physics the magic number is the "Number of nucleons (either protons and nuetrons) such that are arranged into complete shell withen the atomic nucleus.

The seven most widely recognize magic numbers are 2,8,20,28,50,82,126.

The magic nuclei have special stability.

Features of Shell Model

- If neutron (or proton) corresponds to magic number then we need greater energy to remove last neutron(or proton) which is called Separation Energy.
- If both proton and neutron corresponds to magic number then they are most stable nuclei.
- If number of neutrons corresponds to magic number then we have greater number and stable isotones.

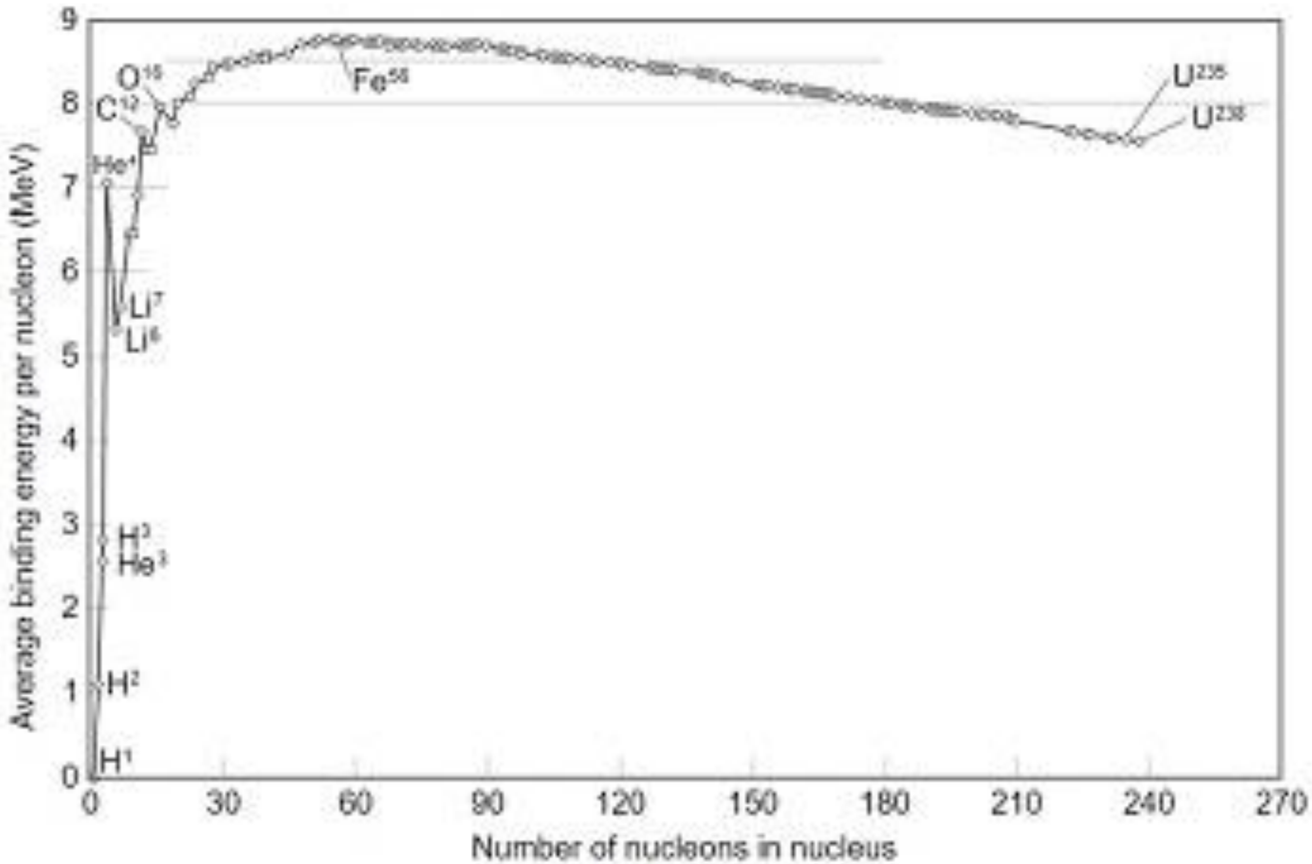
Features of Shell Model

- If number of protons corresponds to magic numbers then we have greater numbers and stable isotopes.

i.e Calcium has six isotopes.

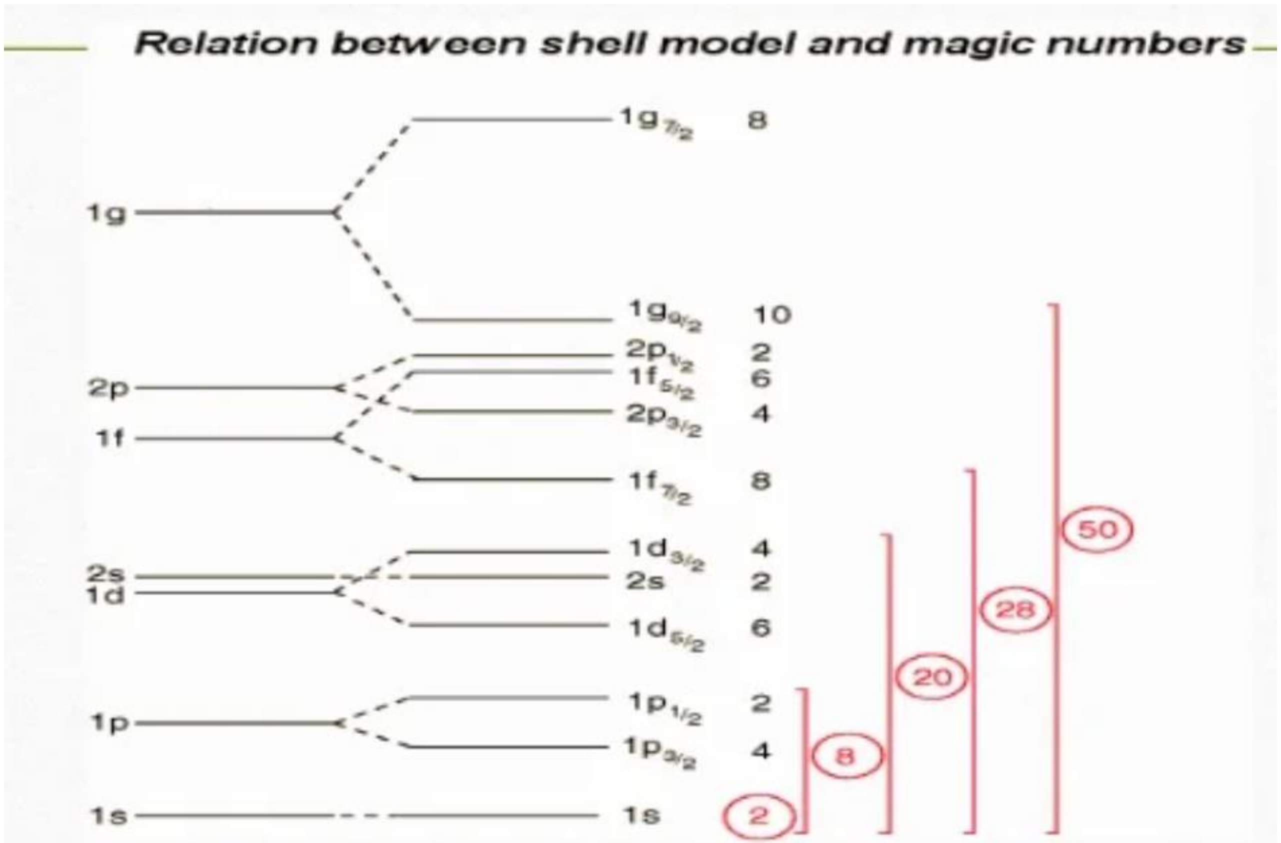
- The element whose Z and N is a magic number has abundance.

B. E Curve



Average binding energy per nucleon (MeV)
Number of nucleons in nucleus
0
1
2
3
4
5
6
7
8
9
0
30
60
90
120
150
180
210
240
270
H1
H2
H3
He3
He4
Li6
Li7
C12
O16
Fe56
U235
U238

Nuclear Shell Structure



Merits of Shell Model

1. It explain the magic number.
2. It explain the magnetic moment of some nuclei.
3. It explain successfully ground state spin.
4. It explain the greater stability and high Binding energy

Limitation of Shell Model

- The first limitation that one notices is that there is a difference between shell-model wave functions and the real state of the nucleus.
- Apart from this, the large value of the quadrupole moment seen in many nuclei is hard to explain using this method.
- The strong-spin orbit interaction cannot be applied using this method.
- The shell model has limited applications and cannot be applied to heavy nuclei.